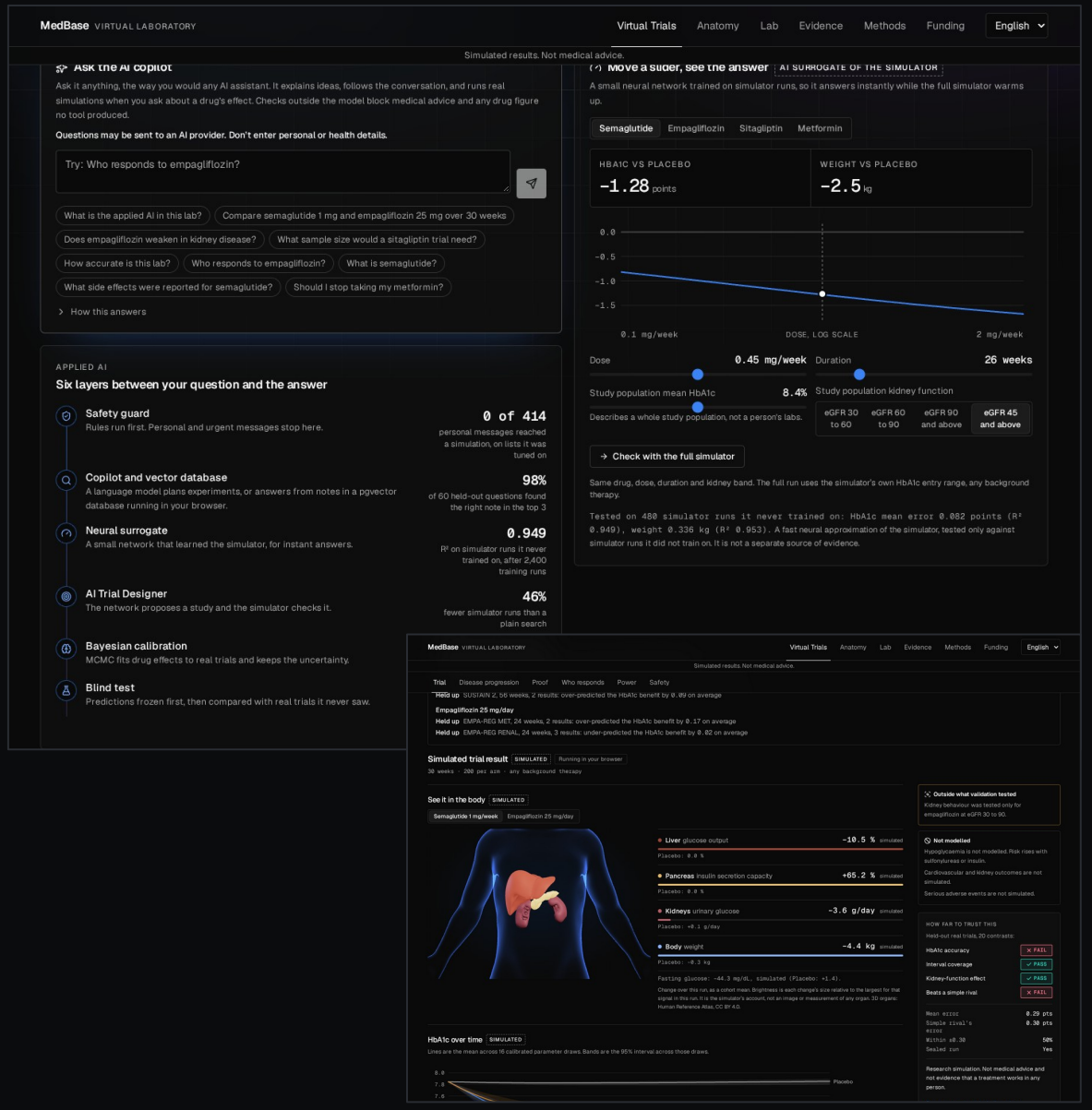
INFINEX 2K26 | APPLIED AI

MedBase Virtual Trials Lab

Rehearse a type 2 diabetes drug trial in your browser, then see how far to trust it.

Try it live
https://zade06079-medbase-virtual-trials-lab.static.hf.space

Simulated results. Not medical advice.



Students can't rehearse a trial, and much health AI can't show where it went wrong

The rehearsal gap

Drug trials cost years and enormous sums. Students and early researchers have no accessible way to explore how a study would behave before it exists, and most open diabetes simulators cover type 1 only.

The trust gap

Many health AI tools produce confident answers with no way to check them. A learner cannot tell a sourced figure from a plausible one, or see where the tool has been wrong before.

Our answer: virtual patients from real survey data, AI that plans and explains, and a simulator that computes every efficacy number.

Our solution: ask in plain English, run a trial, and see how far to trust it

Conversational AI copilot

Explains ideas, follows up on "why?", and runs real simulations when you ask about a drug's effect.

Virtual trials

Four real drugs and hypothetical molecules, on virtual patients built from real health survey labs.

AI Trial Designer

Set a goal. A neural network proposes a study, and the simulator checks it step by step.

Instant neural sliders

Move dose or duration and get an answer in milliseconds, then race it against the full simulator.

Vector database in the browser

PostgreSQL with pgvector finds the right notes by meaning, with no server and no account.

Public blind test

Every frozen prediction beside the real result, shown beside each simulated answer, misses included.

Models propose and explain. The simulator computes.

Bayesian calibration	MCMC learns drug effects from three real trials and carries the uncertainty into every prediction.	**R-hat 1.026**	44 walkers, 6500 steps, 56 autocorrelation times
Neural surrogate	Trained on 2400 simulator runs. Instant sliders, runs in the browser. It learned the simulator, not reality.	**R^2 0.949**	HbA1c error 0.082 points on 480 simulator runs it never saw
AI Trial Designer	The surrogate proposes a dose; the simulator checks it, refines it, then sizes the study.	**46% fewer**	simulator runs than plain search, across 36 goals
Live copilot	A language model on NVIDIA NIM plans tool calls, explains results, and answers ordinary questions in conversation, using notes from a pgvector database in the browser.	**98%**	of 60 held-out questions found the right note in the top 3. Right tool on 10 of 10 experiments
Number and advice guard	A digit figure no tool produced sends the reply back, then falls to computed text. Dose guidance and personal questions are refused.	**44 of 44**	project-written attack prompts refused before any model is called

Technology stack

AI models

- NVIDIA Nemotron 3.5 on NIM: language model
- nemotron-3-embed-1b: embeddings
- Neural surrogate: a small MLP
- Bayesian MCMC with emcee

Science

- Python, NumPy, SciPy
- scikit-learn, statsmodels
- NHANES survey data
- Published trial results

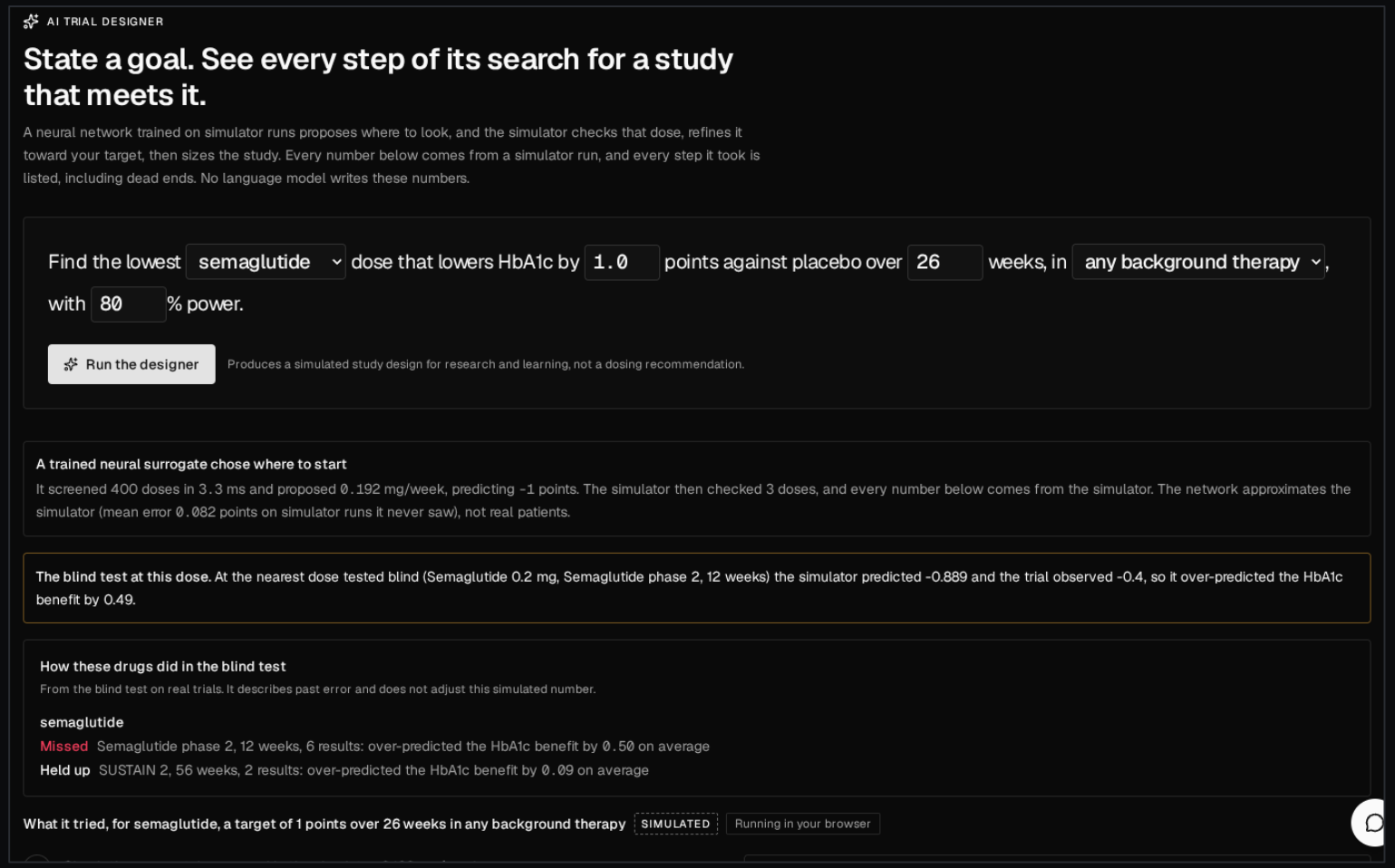
In the browser

- React 19, TypeScript, Vite, Tailwind
- Pyodide: the Python simulator
- PostgreSQL and pgvector 0.8.7, through PGlite
- three.js: 3D body views

Hosting and safety

- Hugging Face Spaces: static site
- Vercel: proxy that holds the AI key
- Rule-based guard before any model
- Content Security Policy, key scans

A neural network chooses where to look, so the simulator runs less



46%

fewer simulator runs: 3.28 on average against 6.03 for plain bisection

36 of 36

goals reached the same verdict on whether a design exists

3%

median dose difference from plain search; largest 8.7%

Testing the copilot found errors a number check can't see, so we changed the design

1

No tool run, no answer

A reply with no tool behind it is rejected. An early draft invented parameter names.

2

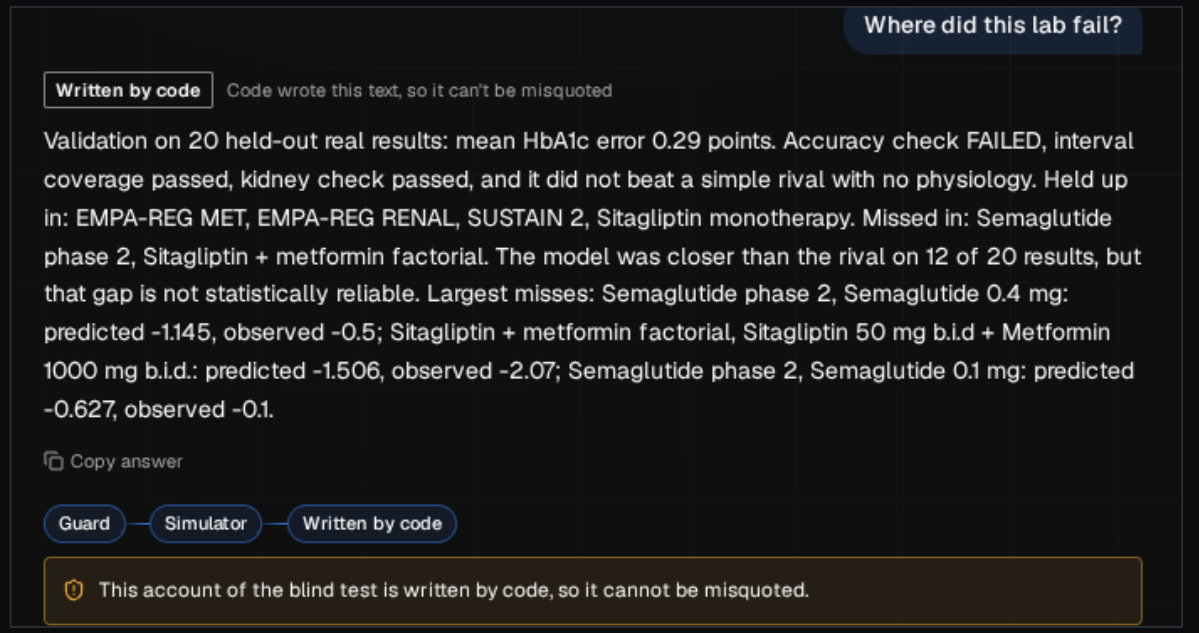
One chance to fix a figure

The guard names the unsourced number and sends the draft back. A second failure loses that line, or the whole draft.

3

Its failures and designs are written by code

A paraphrase got the lab's results backwards, and a design must not read as a dose. The model chooses the tool; code writes the text.



Latest run on 10 project-written questions: 7 answered by the model, 3 written by code, 0 replaced by computed text. Known limit: the guard checks that a figure has a source, not the sentence around it.

We attacked our own safety guard, and it failed before it held

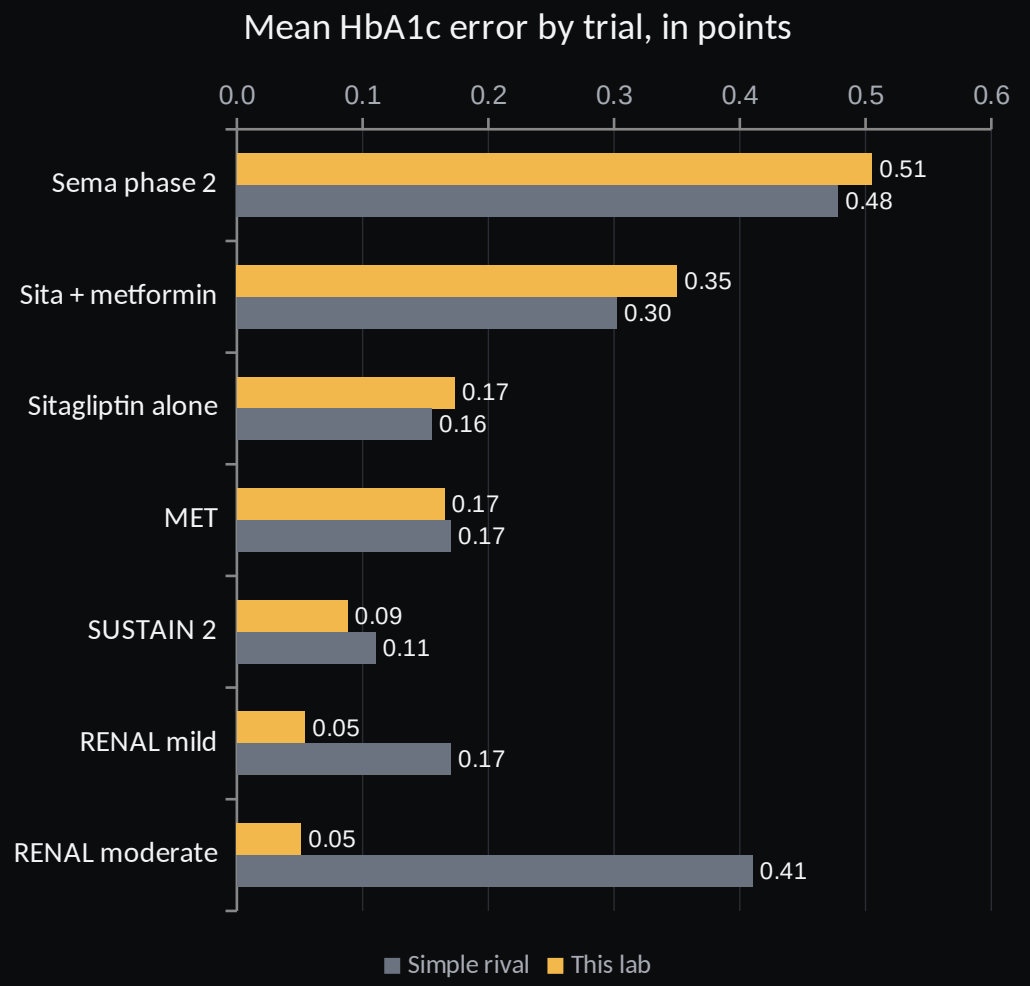
Test	Personal reaching a simulation	Urgent missing the emergency reply	Research questions refused
First attack, after reading the rules	171 got a trial number (201 of 246 wrong class)		
First blind run	0 of 188	17 of 28	15 of 72
Second blind run, terse messages	6 of 226	28 of 58; 4 with no 112	42 of 84
Shipped guard, same lists (tuned, not held out)	0 of 414	1 of 86	18 of 156

- Default is no for simulations: only a research question reaches one. Personal, urgent and non-English messages stop before any model
- Urgent messages are checked first, and every simulation or safety answer ends with 112 and the Tele-MANAS helpline, 14416
- For a supervised research demo, not an unattended public service. Lists written by one author, hard on purpose

Blind test on 20 real results: two passed, two failed

Check	Pass mark	Result	
Accuracy	error ≤ 0.25, 70% within 0.30	error 0.29, 50% within	FAIL
95% interval	≥ 85% of results inside	85%; only 50% inside the 80% interval	PASS
Kidney effect, never fitted	each stratum within 0.30	within, on 3 results	PASS
Beat a rival with no physiology	whole 95% interval below 0	gap -0.01; -0.091 to 0.034	FAIL

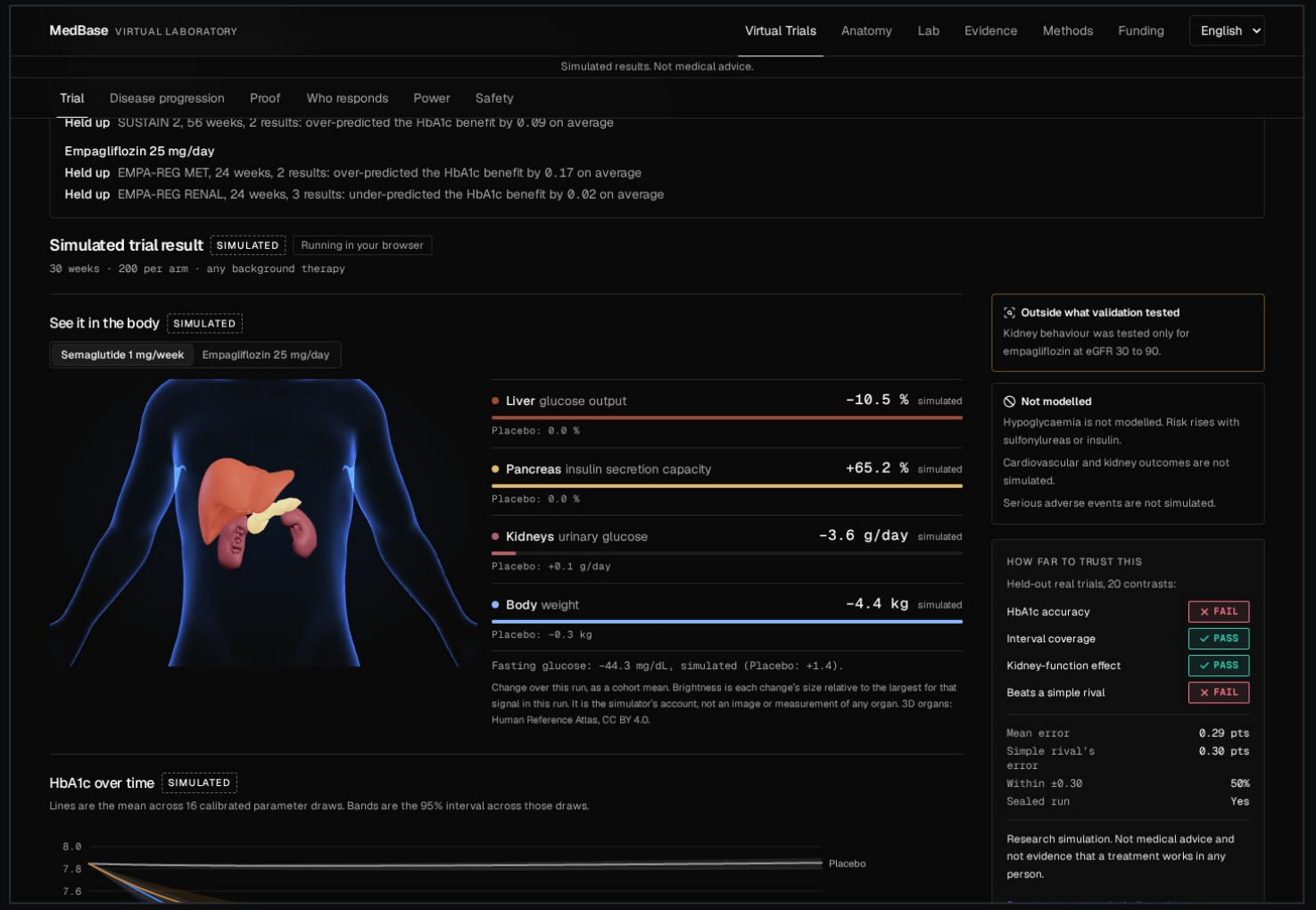
Predictions were frozen and pushed publicly before any result was opened, and results were opened once. The seal is mechanical: these are famous trials, so the authors knew the literature in general.



So the misses stay on screen, beside every simulated result

A learner sees a number and its record together.

- Semaglutide: the 12-week trial missed; the 56-week trial held up
- Empagliflozin: held up in both trial groups, including kidney strata
- The record describes past error and never adjusts the number
- Added after the blind run; it changes no prediction



Innovation and real-world impact

What is new

- A real simulator and a real vector database run inside the browser, at no server cost
- The language model plans and explains, and code checks its figures against what actually ran
- Predictions were frozen before real results were opened, and the failures stay public
- We attacked our own safety guard and published what got through

Who it helps

- Students and early researchers can rehearse a study design for free, with no account, on an ordinary laptop
- Learners see how far to trust an AI answer, because its track record sits beside it
- The AI designer used 46% fewer simulator runs than a plain search, reaching the same verdict
- Next: pilot it with a class, and add Indian patient data

It scales like a static site, and it knows what it is not

Built to spread

- The real simulator runs in the visitor's browser through Pyodide, with no server to pay for
- A small proxy keeps the AI key off the public site
- If the AI is down, an offline planner still answers from computed results

What it is not

- Not clinically validated, and not advice for any person
- No cardiovascular outcomes, hypoglycaemia or serious adverse events simulated
- Virtual patients come from a US survey, not an Indian cohort

Live demo https://zade06079-medbase-virtual-trials-lab.static.hf.space

Code and validation report https://github.com/shreynachaegari-glitch/medbase-virtual-trials-lab

Built on MedBase by Team Aegis.